

# JYOTIRMOY SINGH

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## EDUCATION

**Birla Institute of Technology and Science, Pilani, Hyderabad**

09/2021 - 08/2026

BE (Hons.) Computer Science; MSc (Hons.) Economics | **GPA: 8.6/10**

**Thesis (Advisor: Prof. Anand Rao, Carnegie Mellon University)**

Fall 2025 - Spring 2026

*Multi-Agent AI Orchestration: The System-3/0 Blueprint*

- Designed the System-3/0 Blueprint: a domain-agnostic multi-agent orchestration framework with a System-3 meta-controller and System-0 constrained-decoding safety envelope backed by a Dynamic Agent Registry
- Proved 5 formal results; achieved 3-4× token savings and reduced cascading failure rates from ≈35% to near-zero; extended into MAS-OS with typed IPC and OS-level process isolation.

## EXPERIENCE

**Jovalent** 

Aug 2025 - Present

*Founder & CEO*

*Wyoming, USA*

- Built the full AI stack from scratch: an active-learning OCR pipeline hitting 98% field-level accuracy and a conversational invoice agent at 96% accuracy for automated financial data ingestion.
- Integrated Stripe Connect, QuickBooks, and TaxBandits APIs with async webhooks to automate dunning, W-9 collection, and 1099 generation for US-based SMBs.
- Wrote the confidence-scoring system (field-level penalties + HITL fallback) that keeps error rates under 1% on financial documents in production.

**Google, Google Cloud (Backup for GKE)**

May 2025 - Aug 2025

*Software Engineering Intern*

*Hyderabad, India*

- Wrote the integration test suite for 7 asynchronous schedulers in GKE Backup using Go, gRPC, and GCP internal tooling, taking coverage from 0% to 42%.
- Caught 27+ critical bugs before production rollout through reusable test workflows designed for the scheduler codebase.
- Shipped 3 weeks ahead of schedule and built a generic polling mechanism and custom API clients in Go that replaced flaky async test patterns.

**BITS Pilani x Impactsure** 

Aug 2024 - Dec 2024

*Research Lead (\$150K+ Funded)*

*On-Site*

- Led a team of 4 engineers on a \$150K+ funded project building compact Small Language Models (SLMs) for low-latency deployment at a financial services firm.
- Designed Rail-Only architectures that cut training costs by 20% and infrastructure requirements by 30% compared to standard transformer fine-tuning.

## PUBLICATIONS

**Agentic Anomaly Detection (ORCA)**

Manuscript under Preparation

- Built an edge-deployable anomaly detector combining MinGRU and ConSmax attention with a Gemini-powered Meta-Reasoner for real-time clinical synthesis on MIMIC-IV data.

**Explainable AI for Predictive Healthcare**

Manuscript under Preparation

- Compressed ML ensembles into single interpretable Decision Trees via a Glass-Box distillation pipeline, retaining 96% accuracy on MIMIC-IV clinical data.

## Honors & Recognitions

- Invited to **OpenAI Emerging Talent Community** (2025 Cohort)
- Invited to **Y Combinator** Startup School (In-Person, Bangalore)
- Teaching Assistant, Machine Learning, BITS Pilani (Spring 2025)
- National Level Gold Medalist** in Mathematics

## TECHNICAL SKILLS

**Languages:** Python, Go, C++, Java, SQL

**AI/ML:** PyTorch, TensorFlow, LangChain, LangGraph, Ollama, SHAP, RAG, Multi-Agent Systems

**DevOps & Systems:** Docker, Kubernetes, gRPC, Protocol Buffers, Microservices, Distributed Systems

**Cloud & APIs:** GCP, Stripe Connect, QuickBooks API, Webhooks, REST/GraphQL